

A Proof of Proposition 4.1

PROOF.

- First we define the indicator function as follows:

$$I_j'(j) = \begin{cases} 1, & \text{if item } j \text{ appears in } h \text{ samples drawn without replacement from } \mathbf{P}_k^t(\alpha) \\ 0, & \text{otherwise} \end{cases} \quad (17)$$

Based on Lemma E.1, we have $\mathbb{E}[I_j'(j)] = hp^t(s_j^t, \mathbf{v}_j; \alpha)$, then Equation 6 admits the following expectation representation:

$$\begin{aligned} \mathbf{u}_i(t+1) &= \mathbf{u}_i(t) + \frac{\eta}{h} \mathbb{E} \left[\sum_{j=1}^m I_j'(j) \mathbf{w}_j'(j) \mathbf{v}_j \right] = \mathbf{u}_i(t) + \frac{\eta}{h} \sum_{j=1}^m \mathbb{E}[I_j'(j)] \mathbb{E}[\mathbf{w}_j'(j) \mathbf{v}_j] \\ &= \mathbf{u}_i(t) + \frac{\eta}{h} \sum_{j=1}^m hp^t(s_j^t, \mathbf{v}_j; \alpha) (\mathbf{p}_{\text{pos}}^t(\mathbf{u}_i, \mathbf{v}_{j_1}; \beta, \epsilon) - \mathbf{p}_{\text{neg}}^t(\mathbf{u}_i, \mathbf{v}_{j_2}; \beta, \epsilon)) \mathbf{v}_j = \mathbf{u}_i(t) + \eta \sum_{j=1}^m p^t(s_j^t, \mathbf{v}_j; \alpha) g^t(\mathbf{u}_i, \mathbf{v}_j; \beta, \epsilon) \mathbf{v}_j. \end{aligned} \quad (18)$$

- First we have $\sum_{j=1}^m \mathbf{v}_j \mathbf{v}_j^T = \mathbf{V} \mathbf{V}^T$, $\sum_{j=1}^m \mathbf{v}_j \sum_{k=1}^m \mathbf{v}_k^T = \mathbf{V} \mathbf{1}_{m \times m} \mathbf{V}^T$,

$$\begin{aligned} p^t(s_j^t, \mathbf{v}_j; \alpha) &= \frac{\exp(\alpha \mathbf{v}_j^T s_j^t(t))}{\sum_{k=1}^m \exp(\alpha \mathbf{v}_k^T s_j^t(t))} = \frac{1 + \alpha \mathbf{v}_j^T s_j^t(t) + O(\alpha^2 (\mathbf{v}_j^T s_j^t(t))^2)}{\sum_{k=1}^m (1 + \alpha \mathbf{v}_k^T s_j^t(t) + O(\alpha^2 (\mathbf{v}_k^T s_j^t(t))^2))} \\ &\approx \frac{1 + \alpha \mathbf{v}_j^T s_j^t(t)}{m \left(1 + \frac{\alpha}{m} \sum_{k=1}^m \mathbf{v}_k^T s_j^t(t) \right)} \approx \frac{1}{m} \left(1 + \alpha \mathbf{v}_j^T s_j^t(t) - \frac{\alpha}{m} \sum_{k=1}^m \mathbf{v}_k^T s_j^t(t) \right). \end{aligned} \quad (19)$$

$$g^t(\mathbf{u}_i, \mathbf{v}_j; \beta, \epsilon) = \frac{(1 + \mathbf{v}_j^T \mathbf{u}_i(t))^\beta - (1 - \mathbf{v}_j^T \mathbf{u}_i(t))^\beta}{(1 + \mathbf{v}_j^T \mathbf{u}_i(t))^\beta + (1 - \mathbf{v}_j^T \mathbf{u}_i(t))^\beta} + \epsilon = \frac{(1 + \beta \mathbf{v}_j^T \mathbf{u}_i(t)) - (1 - \beta \mathbf{v}_j^T \mathbf{u}_i(t))}{(1 + \beta \mathbf{v}_j^T \mathbf{u}_i(t)) + (1 - \beta \mathbf{v}_j^T \mathbf{u}_i(t))} + \epsilon = \beta \mathbf{v}_j^T \mathbf{u}_i(t) + \epsilon. \quad (20)$$

Then Equation 18 equals:

$$\begin{aligned} \mathbf{u}_i(t+1) &\approx \mathbf{u}_i(t) + \frac{\eta}{m} \sum_{j=1}^m (1 + \alpha \mathbf{v}_j^T s_j^t(t) - \frac{\alpha}{m} \sum_{k=1}^m \mathbf{v}_k^T s_j^t(t)) (\beta \mathbf{v}_j^T \mathbf{u}_i(t) + \epsilon) \mathbf{v}_j = \mathbf{u}_i(t) + \frac{\eta}{m} \sum_{j=1}^m (\alpha \epsilon \mathbf{v}_j^T s_j^t(t) - \frac{\alpha \epsilon}{m} \sum_{k=1}^m \mathbf{v}_k^T s_j^t(t) + \beta \mathbf{v}_j^T \mathbf{u}_i(t) + \epsilon) \mathbf{v}_j \\ &= \mathbf{u}_i(t) + \frac{\eta}{m} \sum_{j=1}^m ((\alpha \epsilon \mathbf{v}_j^T - \frac{\alpha \epsilon}{m} \sum_{k=1}^m \mathbf{v}_k^T) (\gamma \mathbf{u}_i(t) + (1 - \gamma) \frac{\sum_{j \in N_i} \mathbf{u}_j(t)}{|N_i|}) + \beta \mathbf{v}_j^T \mathbf{u}_i(t) + \epsilon) \mathbf{v}_j \\ &= \frac{\eta \epsilon}{m} \sum_{j=1}^m \mathbf{v}_j + (\mathbf{I}_n + \frac{\eta(\alpha \epsilon \gamma + \beta)}{m} \sum_{j=1}^m \mathbf{v}_j \mathbf{v}_j^T - \frac{\eta \alpha \epsilon \gamma}{m^2} \sum_{j=1}^m \sum_{k=1}^m \mathbf{v}_k \mathbf{v}_k^T) \mathbf{u}_i(t) + (\frac{\eta \alpha \epsilon (1 - \gamma)}{m} \sum_{j=1}^m \mathbf{v}_j \mathbf{v}_j^T - \frac{\eta \alpha \epsilon (1 - \gamma)}{m^2} \sum_{j=1}^m \sum_{k=1}^m \mathbf{v}_k \mathbf{v}_k^T) \frac{\sum_{j \in N_i} \mathbf{u}_j(t)}{|N_i|} \\ &= \frac{\eta \epsilon}{m} \sum_{j=1}^m \mathbf{v}_j + (\mathbf{I}_n + \frac{\eta(\alpha \epsilon \gamma + \beta)}{m} \mathbf{V} \mathbf{V}^T - \frac{\eta \alpha \epsilon \gamma}{m^2} \mathbf{V} \mathbf{1}_{m \times m} \mathbf{V}^T) \mathbf{u}_i(t) + (\frac{\eta \alpha \epsilon (1 - \gamma)}{m} \mathbf{V} \mathbf{V}^T - \frac{\eta \alpha \epsilon (1 - \gamma)}{m^2} \mathbf{V} \mathbf{1}_{m \times m} \mathbf{V}^T) \frac{\sum_{j \in N_i} \mathbf{u}_j(t)}{|N_i|} \end{aligned} \quad (21)$$

Under the definitions of $\mathbf{X}$, $\mathbf{Y}$, and $\mathbf{Z}$ given in Proposition 4.1, we have:

$$\begin{aligned} [\mathbf{u}_i(t+1), \mathbf{u}_2(t+1), \dots, \mathbf{u}_n(t+1)] &= \frac{\eta \epsilon}{m} \sum_{j=1}^m \mathbf{v}_j + \sum_{j=1}^m \mathbf{v}_j \sum_{j_1=1}^m \mathbf{v}_{j_1}^T + \mathbf{Y} [\mathbf{u}_i(t), \mathbf{u}_2(t), \dots, \mathbf{u}_n(t)] + \mathbf{Z} [\frac{\sum_{j \in N_i} \mathbf{u}_j(t)}{|N_i|}, \frac{\sum_{j \in N_2} \mathbf{u}_j(t)}{|N_2|}, \dots, \frac{\sum_{j \in N_n} \mathbf{u}_j(t)}{|N_n|}] \\ \mathbf{U}(t+1) &= \frac{\eta \epsilon}{m} \mathbf{V} \mathbf{1}_{m \times n} + \mathbf{Y} \mathbf{U}(t) + \mathbf{Z} \mathbf{U}(t) (\text{diag}(\mathbf{S} \mathbf{1}_{n \times 1})^{-1} \mathbf{S})^T = \mathbf{X} + \mathbf{Y} \mathbf{U}(t) + \mathbf{Z} \mathbf{U}(t) \mathbf{S}^T \end{aligned} \quad (22)$$

□

B Proof of Proposition 4.2

PROOF.

Under the column-wise vectorization operator $\text{vec}(\cdot): \mathbb{R}^{n \times m} \rightarrow \mathbb{R}^{nm}$, Equation 8 transforms to:

$$\text{vec}(\mathbf{U}(t+1)) = \text{vec}(\mathbf{X}) + \text{vec}(\mathbf{Y} \mathbf{U}(t)) + \text{vec}(\mathbf{Z} \mathbf{U}(t) \mathbf{S}^T). \quad (23)$$

Based on Lemma E.2, we have

$$\text{vec}(\mathbf{Y} \mathbf{U}(t)) = \text{vec}(\mathbf{Y} \mathbf{U}(t) \mathbf{I}_n^T) = (\mathbf{I}_n \otimes \mathbf{Y}) \text{vec}(\mathbf{U}(t)); \quad \text{vec}(\mathbf{Z} \mathbf{U}(t) \mathbf{S}^T) = (\mathbf{S} \otimes \mathbf{Z}) \text{vec}(\mathbf{U}(t)). \quad (24)$$

$$\text{vec}(\mathbf{U}(t+1)) = \text{vec}(\mathbf{X}) + (\mathbf{I}_n \otimes \mathbf{Y}) \text{vec}(\mathbf{U}(t)) + (\mathbf{S} \otimes \mathbf{Z}) \text{vec}(\mathbf{U}(t)) = \text{vec}(\mathbf{X}) + (\mathbf{I}_n \otimes \mathbf{Y} + \mathbf{S} \otimes \mathbf{Z}) \text{vec}(\mathbf{U}(t)). \quad (25)$$

Assume that $\mathbf{U}(t)$ in Equation 25 converges to $\mathbf{U}^*$, we have

$$(\mathbf{I}_{nc} - (\mathbf{I}_n \otimes \mathbf{Y} + \mathbf{S} \otimes \mathbf{Z})) \text{vec}(\mathbf{U}^*) = \text{vec}(\mathbf{X}) \quad (26)$$

Now, the convergence of Equation 8 is equivalent to the invertibility of $(\mathbf{I}_{nc} - (\mathbf{I}_n \otimes \mathbf{Y} + \mathbf{S} \otimes \mathbf{Z}))$. To establish that $\eta(\frac{\epsilon}{2} + \frac{\alpha \epsilon \gamma}{2} + \frac{\beta \epsilon}{\alpha \epsilon \gamma}) < 1$ is sufficient for $(\mathbf{I}_{nc} - (\mathbf{I}_n \otimes \mathbf{Y} + \mathbf{S} \otimes \mathbf{Z}))^{-1}$ exists, we proceed through a sequence of logical implications:

- We first demonstrate that $\eta(\frac{\epsilon}{2} + \frac{\alpha \epsilon \gamma}{2} + \frac{\beta \epsilon}{\alpha \epsilon \gamma}) < 1 \Rightarrow \|\mathbf{Y}\|_\infty + \|\mathbf{Z}\|_\infty < 1$;

Proof. We previously defined the representation of item j as $\mathbf{v}_j = [v_j^{(1)}, v_j^{(2)}, \dots, v_j^{(c)}]^T$, $j \in \{1, 2, \dots, m\}$, if item j belongs to category k , then $v_j^{(p)} v_j^{(q)} = 1, v_j^{(p)} v_j^{(q)} = 0$ for $p \neq k$ or $q \neq k$, and $\sum_{j=1}^m v_j^{(q)} = n_k, \sum_{j=1}^m v_j^{(p)} = n_k$.

$$\mathbf{V} \mathbf{V}^T = \sum_{j=1}^m \mathbf{v}_j \mathbf{v}_j^T = \begin{bmatrix} \sum_{j=1}^m v_j^{(1)} v_j^{(1)} & \dots & \sum_{j=1}^m v_j^{(1)} v_j^{(c)} \\ \vdots & \ddots & \vdots \\ \sum_{j=1}^m v_j^{(c)} v_j^{(1)} & \dots & \sum_{j=1}^m v_j^{(c)} v_j^{(c)} \end{bmatrix} = \begin{bmatrix} n_1 & 0 & \dots & 0 \\ 0 & n_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & n_c \end{bmatrix} \quad (27)$$

$$\mathbf{V} \mathbf{1}_{m \times m} \mathbf{V}^T = \sum_{j=1}^m \mathbf{v}_j \sum_{k=1}^m \mathbf{v}_k^T = \begin{bmatrix} \sum_{j=1}^m v_j^{(1)} \sum_{k=1}^m v_k^{(1)} & \dots & \sum_{j=1}^m v_j^{(1)} \sum_{k=1}^m v_k^{(c)} \\ \vdots & \ddots & \vdots \\ \sum_{j=1}^m v_j^{(c)} \sum_{k=1}^m v_k^{(1)} & \dots & \sum_{j=1}^m v_j^{(c)} \sum_{k=1}^m v_k^{(c)} \end{bmatrix} = \begin{bmatrix} n_1 \cdot n_1 & \dots & n_1 \cdot n_c \\ \vdots & \ddots & \vdots \\ n_c \cdot n_1 & \dots & n_c \cdot n_c \end{bmatrix} \quad (28)$$

$$\mathbf{Y} \approx \frac{\eta(\alpha \epsilon \gamma + \beta)}{m} \begin{bmatrix} n_1 & 0 & \dots & 0 \\ 0 & n_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & n_c \end{bmatrix} - \frac{\eta \alpha \epsilon \gamma}{m^2} \begin{bmatrix} n_1 \cdot n_1 & \dots & n_1 \cdot n_c \\ \vdots & \ddots & \vdots \\ n_c \cdot n_1 & \dots & n_c \cdot n_c \end{bmatrix} \quad (29)$$

$$\mathbf{Z} = \frac{\eta \alpha \epsilon (1 - \gamma)}{m} \left(\begin{bmatrix} n_1 & 0 & \dots & 0 \\ 0 & n_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & n_c \end{bmatrix} - \frac{1}{m} \begin{bmatrix} n_1 \cdot n_1 & \dots & n_1 \cdot n_c \\ \vdots & \ddots & \vdots \\ n_c \cdot n_1 & \dots & n_c \cdot n_c \end{bmatrix} \right) \quad (30)$$

$$\|\mathbf{Y}\|_\infty = \max_{1 \leq i \leq c} \left(\frac{\eta \alpha \epsilon \gamma}{m^2} n_i \left(\sum_{j=1}^c n_j \right) - \frac{\eta \alpha \epsilon \gamma}{m^2} n_i^2 + \frac{\eta(\alpha \epsilon \gamma + \beta)}{m} n_i - \frac{\eta \alpha \epsilon \gamma}{m^2} n_i^2 \right) = \max_{1 \leq i \leq c} \left(\frac{\eta(2 \alpha \epsilon \gamma + \beta)}{m} n_i - \frac{2 \eta \alpha \epsilon \gamma}{m^2} n_i^2 \right) \leq \eta \frac{(2 \alpha \epsilon \gamma + \beta)^2}{8 \alpha \epsilon \gamma} \quad (31)$$

$$\|\mathbf{Z}\|_\infty = \frac{\eta \alpha \epsilon (1 - \gamma)}{m} \max_{1 \leq i \leq c} \left(\frac{1}{m} n_i \left(\sum_{j=1}^c n_j \right) - \frac{1}{m} n_i^2 + n_i - \frac{1}{m} n_i^2 \right) = \frac{2 \eta \alpha \epsilon (1 - \gamma)}{m} \max_{1 \leq i \leq c} \left(n_i - \frac{1}{m} n_i^2 \right) \leq \frac{2 \eta \alpha \epsilon (1 - \gamma)}{m} \cdot \frac{m}{4} = \eta \frac{\alpha \epsilon (1 - \gamma)}{2} \quad (32)$$

Then $\|\mathbf{Y}\|_\infty + \|\mathbf{Z}\|_\infty \leq \eta \frac{(2 \alpha \epsilon \gamma + \beta)^2}{4} + \eta \frac{\alpha \epsilon (1 - \gamma)}{2} = \eta \left(\frac{\epsilon}{2} + \frac{\alpha \epsilon \gamma}{2} + \frac{\beta \epsilon}{\alpha \epsilon \gamma} \right) < 1$. □

- Next, we demonstrate that $\|\mathbf{Y}\|_\infty + \|\mathbf{Z}\|_\infty < 1 \Rightarrow \|\mathbf{I}_n \otimes \mathbf{Y} + \mathbf{S} \otimes \mathbf{Z}\|_\infty < 1$. Proof. Given the triangle inequality property of norms and Lemma E.6, we derive:

$$\|\mathbf{I}_n \otimes \mathbf{Y} + \mathbf{S} \otimes \mathbf{Z}\|_\infty \leq \|\mathbf{I}_n \otimes \mathbf{Y}\|_\infty + \|\mathbf{S} \otimes \mathbf{Z}\|_\infty = \|\mathbf{I}_n\|_\infty \|\mathbf{Y}\|_\infty + \|\mathbf{S}\|_\infty \|\mathbf{Z}\|_\infty$$

Furthermore, $\|\mathbf{I}_n\|_\infty = 1$, and $\mathbf{S} = \text{diag}(\mathbf{S} \mathbf{1}_{n \times 1})^{-1} \mathbf{S}$, then $\forall k, \sum_i |\mathbf{S}_{ki}| = 1, \|\mathbf{S}\|_\infty = 1$. Consequently, we obtain $\|\mathbf{I}_n \otimes \mathbf{Y} + \mathbf{S} \otimes \mathbf{Z}\|_\infty \leq \|\mathbf{Y}\|_\infty + \|\mathbf{Z}\|_\infty < 1$. □

- Finally, based on Lemma E.5, we have $\|\mathbf{I}_n \otimes \mathbf{Y} + \mathbf{S} \otimes \mathbf{Z}\|_\infty < 1 \Rightarrow (\mathbf{I}_{nc} - (\mathbf{I}_n \otimes \mathbf{Y} + \mathbf{S} \otimes \mathbf{Z}))^{-1}$ exists.

Therefore, we prove that $\eta(\frac{\epsilon}{2} + \frac{\alpha \epsilon \gamma}{2} + \frac{\beta \epsilon}{\alpha \epsilon \gamma}) < 1$ is sufficient for $(\mathbf{I}_{nc} - (\mathbf{I}_n \otimes \mathbf{Y} + \mathbf{S} \otimes \mathbf{Z}))^{-1}$ exists, i.e., the $\mathbf{U}(t)$ of Equation 8 converges to $\mathbf{U}^* = \text{unvec}(\mathbf{I}_{nc} - (\mathbf{I}_n \otimes \mathbf{Y} + \mathbf{S} \otimes \mathbf{Z}))^{-1} \text{vec}(\mathbf{X})$. □

C Proof of COROLLARY 4.3

PROOF.

- Transient Homogenization
When $\epsilon = 0, \gamma = 1$, Equation 21 reduces to

$$\mathbf{u}_i(t+1) = (\mathbf{I}_n + \frac{\eta \beta}{m} \mathbf{V} \mathbf{V}^T) \mathbf{u}_i(t), \quad (33)$$

Let $\lambda = \frac{\eta \beta}{m}$, and applying Equation 27, we have

$$\begin{bmatrix} \mathbf{u}_i^{(1)}(t+1) \\ \mathbf{u}_i^{(2)}(t+1) \\ \vdots \\ \mathbf{u}_i^{(c)}(t+1) \end{bmatrix} = \begin{bmatrix} 1 + \lambda n_1 & 0 & \dots & 0 \\ 0 & 1 + \lambda n_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 + \lambda n_c \end{bmatrix} \begin{bmatrix} \mathbf{u}_i^{(1)}(t) \\ \mathbf{u}_i^{(2)}(t) \\ \vdots \\ \mathbf{u}_i^{(c)}(t) \end{bmatrix} = \begin{bmatrix} (1 + \lambda n_1) \mathbf{u}_i^{(1)}(t) \\ (1 + \lambda n_2) \mathbf{u}_i^{(2)}(t) \\ \vdots \\ (1 + \lambda n_c) \mathbf{u}_i^{(c)}(t) \end{bmatrix} \quad (34)$$

If $\exists k$, s.t. $\mathbf{u}_i^{(k)}(t) \mathbf{u}_j^{(k)}(t) \geq \frac{\|\mathbf{u}_i(t)\|_2 \|\mathbf{u}_j(t)\|_2}{\sqrt{1 + (\frac{2 n_k + \lambda n_k^2}{\sum_{l \neq k} 2 n_l + \lambda n_l^2})^2}}$, based on Lemma E.7, we have

$$\min_{i \neq k} \mathbf{u}_i^{(i)}(t) \mathbf{u}_j^{(i)}(t) \geq - \sqrt{\frac{\|\mathbf{u}_i(t)\|_2^2 \|\mathbf{u}_j(t)\|_2^2}{1 + (\frac{2 n_k + \lambda n_k^2}{\sum_{l \neq k} 2 n_l + \lambda n_l^2})^2}} - \frac{\frac{\|\mathbf{u}_i(t)\|_2 \|\mathbf{u}_j(t)\|_2}{\sqrt{1 + (\frac{2 n_k + \lambda n_k^2}{\sum_{l \neq k} 2 n_l + \lambda n_l^2})^2}}}{\sum_{l \neq k} (2 n_l + \lambda n_l^2) \sqrt{1 + (\frac{2 n_l + \lambda n_l^2}{\sum_{m \neq l} 2 n_m + \lambda n_m^2})^2}}} \quad (35)$$

Then

$$\begin{aligned} \mathbf{u}_i^T(t+1) \mathbf{u}_j(t+1) - \mathbf{u}_i^T(t) \mathbf{u}_j(t) &= \sum_{l=1}^c (2 \lambda n_l + \lambda^2 n_l^2) \mathbf{u}_i^{(l)}(t) \mathbf{u}_j^{(l)}(t) \\ &= \lambda \left((2 n_k + \lambda n_k^2) \mathbf{u}_i^{(k)}(t) \mathbf{u}_j^{(k)}(t) + \sum_{l \neq k} (2 n_l + \lambda n_l^2) \mathbf{u}_i^{(l)}(t) \mathbf{u}_j^{(l)}(t) \right) \geq \lambda \left((2 n_k + \lambda n_k^2) \mathbf{u}_i^{(k)}(t) \mathbf{u}_j^{(k)}(t) + \sum_{l \neq k} (2 n_l + \lambda n_l^2) \min_{i \neq k} \mathbf{u}_i^{(l)}(t) \mathbf{u}_j^{(l)}(t) \right) \\ &\geq \lambda \left((2 n_k + \lambda n_k^2) \frac{\|\mathbf{u}_i(t)\|_2 \|\mathbf{u}_j(t)\|_2}{\sqrt{1 + (\frac{2 n_k + \lambda n_k^2}{\sum_{l \neq k} 2 n_l + \lambda n_l^2})^2}} - \sum_{l \neq k} (2 n_l + \lambda n_l^2) \frac{\|\mathbf{u}_i(t)\|_2 \|\mathbf{u}_j(t)\|_2}{\sum_{l \neq k} (2 n_l + \lambda n_l^2) \sqrt{1 + (\frac{2 n_l + \lambda n_l^2}{\sum_{m \neq l} 2 n_m + \lambda n_m^2})^2}} \right) = 0 \end{aligned} \quad (36)$$

- Steady-state Homogenization

$$\text{Let } \mathbf{D} = \begin{bmatrix} 1 + \lambda n_1 & 0 & \dots & 0 \\ 0 & 1 + \lambda n_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 + \lambda n_c \end{bmatrix}, d_i = D_{ii} = 1 + \lambda n_i$$

Proof by Mathematical Induction:

$$\text{If } k = \arg \max_{i \in \{1, 2, \dots, c\}} n_{i_0}, \text{ then } \forall \tau \in \mathbb{Z}^+, \text{ we have } \mathbf{u}_i^{(k)}(t + \tau - 1) \mathbf{u}_j^{(k)}(t + \tau - 1) \geq \frac{\|\mathbf{u}_i(t + \tau - 1)\|_2 \|\mathbf{u}_j(t + \tau - 1)\|_2}{\sqrt{1 + (\frac{2 n_k + \lambda n_k^2}{\sum_{l \neq k} 2 n_l + \lambda n_l^2})^2}}. \quad (37)$$

Base Case: We have verified that the inequality holds when we let $\tau = 1$, i.e.,

$$\mathbf{u}_i^{(k)}(t) \mathbf{u}_j^{(k)}(t) \geq \frac{\|\mathbf{u}_i(t)\|_2 \|\mathbf{u}_j(t)\|_2}{\sqrt{1 + (\frac{2 n_k + \lambda n_k^2}{\sum_{l \neq k} 2 n_l + \lambda n_l^2})^2}}, \quad (38)$$

and we also have $d_k = 1 + \lambda n_k = 1 + \lambda \max_{i \in \{1, 2, \dots, c\}} n_{i_0} = \max_{i \in \{1, 2, \dots, c\}} (1 + \lambda n_{i_0}) = \max_{i \in \{1, 2, \dots, c\}} d_{i_0}$.

Inductive Hypothesis: Assume that the inequality holds for $\tau = m \in \mathbb{Z}^+$, i.e.,

$$\mathbf{u}_i^{(k)}(t + m - 1) \mathbf{u}_j^{(k)}(t + m - 1) \geq \frac{\|\mathbf{u}_i(t + m - 1)\|_2 \|\mathbf{u}_j(t + m - 1)\|_2}{\sqrt{1 + (\frac{2 n_k + \lambda n_k^2}{\sum_{l \neq k} 2 n_l + \lambda n_l^2})^2}}. \quad (39)$$

Inductive Step: When $\tau = m + 1$, based on Lemma E.8, we have

$$\begin{aligned} \frac{\mathbf{u}_i^{(k)}(t + m) \mathbf{u}_j^{(k)}(t + m)}{\|\mathbf{u}_i(t + m)\|_2 \|\mathbf{u}_j(t + m)\|_2} &= \frac{d_k^2 \mathbf{u}_i^{(k)}(t + m - 1) \mathbf{u}_j^{(k)}(t + m - 1)}{\|\mathbf{D} \mathbf{u}_i(t + m - 1)\|_2 \|\mathbf{D} \mathbf{u}_j(t + m - 1)\|_2} \geq \frac{d_k^2 \mathbf{u}_i^{(k)}(t + m - 1) \mathbf{u}_j^{(k)}(t + m - 1)}{(\max_i d_i)^2 \|\mathbf{u}_i(t + m - 1)\|_2 \|\mathbf{u}_j(t + m - 1)\|_2} \\ &\geq \frac{d_k^2 \mathbf{u}_i^{(k)}(t + m - 1) \mathbf{u}_j^{(k)}(t + m - 1)}{d_k^2 \|\mathbf{u}_i(t + m - 1)\|_2 \|\mathbf{u}_j(t + m - 1)\|_2} = \frac{\mathbf{u}_i^{(k)}(t + m - 1) \mathbf{u}_j^{(k)}(t + m - 1)}{\|\mathbf{u}_i(t + m - 1)\|_2 \|\mathbf{u}_j(t + m - 1)\|_2} \geq \frac{1}{\sqrt{1 + (\frac{2 n_k + \lambda n_k^2}{\sum_{l \neq k} 2 n_l + \lambda n_l^2})^2}} \\ &\text{i.e. } \mathbf{u}_i^{(k)}(t + m) \mathbf{u}_j^{(k)}(t + m) \geq \frac{\|\mathbf{u}_i(t + m)\|_2 \|\mathbf{u}_j(t + m)\|_2}{\sqrt{1 + (\frac{2 n_k + \lambda n_k^2}{\sum_{l \neq k} 2 n_l + \lambda n_l^2})^2}}. \end{aligned} \quad (40)$$

After leveraging the inductive hypothesis, we show the inequality continues to hold when $\tau = m + 1$.

Conclusion:

By the principle of mathematical induction, $\forall \tau \in \mathbb{Z}^+$, we have $\mathbf{u}_i^{(k)}(t + \tau - 1) \mathbf{u}_j^{(k)}(t + \tau - 1) \geq \frac{\|\mathbf{u}_i(t + \tau - 1)\|_2 \|\mathbf{u}_j(t + \tau - 1)\|_2}{\sqrt{1 + (\frac{2 n_k + \lambda n_k^2}{\sum_{l \neq k} 2 n_l + \lambda n_l^2})^2}}$, so we also have $\forall \tau \in \mathbb{Z}^+$, $\mathbf{u}_i^T(t + \tau) \mathbf{u}_j(t + \tau) \geq \mathbf{u}_i^T(t + \tau - 1) \mathbf{u}_j(t + \tau - 1)$. □

D Proof of COROLLARY 4.4

PROOF.

From Equation 34: $\forall \alpha, i, \mathbf{u}_i^{(\alpha)}(t+1) = (1 + \lambda n_{\alpha}) \mathbf{u}_i^{(\alpha)}(t)$, and this is equivalent to $\Delta t = 1, \frac{d \mathbf{u}_i^{(\alpha)}(t + \Delta t) - \mathbf{u}_i^{(\alpha)}(t)}{\Delta t} = \lambda n_{\alpha} \mathbf{u}_i^{(\alpha)}(t)$, when $\Delta t \rightarrow 0$, we similarly obtain

$$\frac{d \mathbf{u}_i^{(\alpha)}(t)}{dt} = \frac{\mathbf{u}_i^{(\alpha)}(t + \Delta t) - \mathbf{u}_i^{(\alpha)}(t)}{\Delta t} = \lambda n_{\alpha} \mathbf{u}_i^{(\alpha)}(t) \quad (41)$$

Then

$$P_{i, \text{Lo}}^t := \frac{S_{i, \text{Lo}}^t(t)}{\sum_{k=1}^c S_{i, \text{Lo}}^t(t)} = P(\text{user } i \text{ receives an item from Category } o) = \frac{n_{\alpha} e^{\alpha \mathbf{u}_i^{(\alpha)}(t)}}{\sum_{k=1}^c n_k e^{\alpha \mathbf{u}_i^{(k)}(t)}} \quad (42)$$

The time derivative of recommended categories entropy for user i at time t is given by

$$\frac{d H_i(t)}{dt} = - \sum_{\alpha=1}^c \left(\frac{d P_{i, \text{Lo}}^t}{dt} \ln P_{i, \text{Lo}}^t + P_{i, \text{Lo}}^t \frac{d \ln P_{i, \text{Lo}}^t}{dt} \right) = - \sum_{\alpha=1}^c \ln P_{i, \text{Lo}}^t \frac{d P_{i, \text{Lo}}^t}{dt} - \sum_{\alpha=1}^c \frac{d P_{i, \text{Lo}}^t}{dt} = - \sum_{\alpha=1}^c \ln P_{i, \text{Lo}}^t \frac{d P_{i, \text{Lo}}^t}{dt} - \frac{d \sum_{\alpha=1}^c P_{i, \text{Lo}}^t}{dt} = - \sum_{\alpha=1}^c \ln P_{i, \text{Lo}}^t \frac{d P_{i, \text{Lo}}^t}{dt} \quad (43$$

E Supporting Lemmas

LEMMA E.1.

$$\mathbb{E}[I'_{ik}(j)] \approx h p^i(s'_j, v_j; \alpha) \quad (47)$$

PROOF.

First we define the selection indicator:

$$I'_{ik}(j) = \begin{cases} 1, & \text{if item } j \text{ is selected at the } k\text{-th draw without replacement from } p'_i(\alpha) \\ 0, & \text{otherwise} \end{cases} \quad (48)$$

For simplicity, we denote $p^i(s'_j, v_j; \alpha)$ as p_j , because of $h \ll m, p_j \rightarrow 0$, we have:

$$\mathbb{E}[I'_{ik}(j)] = \sum_{k=1}^h \mathbb{E}[I'_{ik}(j)] \approx \sum_{k=1}^h p_j = h p_j \quad (49)$$

Then we compute the error:

$$\delta = \sum_{k=1}^h p_j - \sum_{k=1}^h \mathbb{E}[I'_{ik}(j)] \approx h p_j - (p_j + (h-1) \sum_{l \neq j} p_l p_j) = h p_j - (p_j + (h-1)(1-p_j)p_j) = (h-1)p_j^2 = (h-1)(p^i(s'_j, v_j; \alpha))^2 \quad (50)$$

□

LEMMA E.2.

$$\text{vec}(AUB^T) = (B \otimes A) \text{vec}(U) \quad (51)$$

where $A \in \mathbb{R}^{n \times n}$, $U \in \mathbb{R}^{n \times n}$, $B \in \mathbb{R}^{m \times n}$.

PROOF.

$\forall i, j$, $(AUB^T)_{ij} = \sum_{k=1}^n A_{ik} U_{kj} B_{kj}^T$ is equal to the $(j-1)c + i$ -th row element of matrix $\text{vec}(AUB^T)$.

Furthermore,

$$B \otimes A = \begin{bmatrix} B_{11}A & \dots & B_{1n}A \\ \vdots & \ddots & \vdots \\ B_{n1}A & \dots & B_{nn}A \end{bmatrix} \in \mathbb{R}^{m \times n \times n} \quad (52)$$

the $(j-1)c + i$ -th row element of matrix $(B \otimes A) \text{vec}(U)$ is equal to $\sum_{k=1}^n \sum_{l=1}^m A_{ik} U_{lj} B_{lj}^T$. Therefore, all corresponding elements of $\text{vec}(AUB^T)$ and $(B \otimes A) \text{vec}(U)$ are equal, i.e., $\text{vec}(AUB^T) = (B \otimes A) \text{vec}(U)$. □

LEMMA E.3.

$$\|AB\|_\infty \leq \|A\|_\infty \|B\|_\infty \quad (53)$$

PROOF.

For a matrix $A \in \mathbb{R}_{m \times n}$, the infinity norm is defined as $\|A\|_\infty = \max_{1 \leq i \leq m} \sum_{j=1}^n |A_{ij}|$. Since $\sum_j |B_{ji}| \leq \max_j \sum_i |B_{ji}| = \|B\|_\infty$, $\sum_j |A_{ij}| \leq \max_k \sum_j |A_{kj}| = \|A\|_\infty$.

$$\text{Then } \forall k, \sum_i |(AB)_{ki}| = \sum_i \left| \sum_j A_{kj} B_{ji} \right| \leq \sum_j |A_{kj}| \sum_i |B_{ji}| = \sum_j |A_{kj}| \sum_i |B_{ji}| \leq \sum_j |A_{kj}| \|B\|_\infty \leq \|A\|_\infty \|B\|_\infty \quad (54)$$

Therefore $\|AB\|_\infty = \max_k \sum_i |(AB)_{ki}| \leq \|A\|_\infty \|B\|_\infty$. □

LEMMA E.4.

$$\rho(A) \leq \|A\|_\infty \quad (55)$$

where $\rho(A)$ denotes the spectral radius of matrix A .

PROOF.

Let λ be an eigenvalue of matrix A , and let x be a corresponding eigenvector such that $Ax = \lambda x$ and $x \neq 0$. Then $\|Ax\|_\infty = \|\lambda x\|_\infty = |\lambda| \|x\|_\infty$. Since $\|Ax\|_\infty \leq \|A\|_\infty \|x\|_\infty$ based on Lemma E.3, we get $|\lambda| \|x\|_\infty \leq \|A\|_\infty \|x\|_\infty$. And $\|x\|_\infty > 0$ with $x \neq 0$, we have $|\lambda| \leq \|A\|_\infty$, this inequality holds for all eigenvalues λ of matrix A . Therefore, $\rho(A) = \max\{|\lambda| \mid \lambda \text{ eigenvalues of } A\} \leq \|A\|_\infty$. □

LEMMA E.5.

A sufficient condition for the invertibility of $I - A$ is that the infinity norm of A is strictly less than 1, i.e.,

$$\|A\|_\infty < 1 \Rightarrow (I - A)^{-1} \text{ exists.} \quad (57)$$

PROOF.

Based on Lemma E.4, we get $\rho(A) \leq \|A\|_\infty$. Given that $\|A\|_\infty < 1$, it follows that $\rho(A) < 1$. This implies that all eigenvalues λ of A satisfy $|\lambda| < 1$, i.e., all eigenvalues of $I - A$ satisfy $\mu = 1 - \lambda > 0$. Hence, $I - A$ has no zero eigenvalues, which implies that $\det(I - A) \neq 0$. Therefore, $I - A$ is invertible. □

LEMMA E.6.

$$\|A \otimes B\|_\infty = \|A\|_\infty \|B\|_\infty \quad (58)$$

PROOF.

Since

$$\|A\|_\infty = \max_i \sum_j |A_{ij}|, \|B\|_\infty = \max_k \sum_l |B_{kl}|. \quad (59)$$

Then

$$\|A \otimes B\|_\infty = \max_{ik} \sum_{jl} |A_{ij} B_{kl}| = \max_i \sum_j |A_{ij}| \cdot \max_k \sum_l |B_{kl}| = \|A\|_\infty \|B\|_\infty \quad (60)$$

□

LEMMA E.7.

Let $a = [a^{(1)}, a^{(2)}, \dots, a^{(n)}]^T \in \mathbb{R}^n$, $b = [b^{(1)}, b^{(2)}, \dots, b^{(n)}]^T \in \mathbb{R}^n$.

If there exists an index $k \in \{1, \dots, n\}$ such that $a^{(k)} b^{(k)} \geq x > 0$, then a lower bound for the remaining products is given by

$$\min_{i \neq k} a^{(i)} b^{(i)} \geq -\sqrt{\|a\|_2^2 \|b\|_2^2 - x^2}. \quad (61)$$

PROOF.

First

$$\sum_i (a^{(i)} b^{(i)})^2 = \sum_i (a^{(i)})^2 (b^{(i)})^2 \leq \sum_i (a^{(i)})^2 \sum_i (b^{(i)})^2 = \|a\|_2^2 \|b\|_2^2 \quad (62)$$

Given $a^{(k)} b^{(k)} \geq x > 0$,

$$\sum_{i \neq k} (a^{(i)} b^{(i)})^2 \leq \|a\|_2^2 \|b\|_2^2 - x^2. \quad (63)$$

This implies that for all $i \neq k$,

$$|a^{(i)} b^{(i)}| \leq \sqrt{\|a\|_2^2 \|b\|_2^2 - x^2}, \quad (64)$$

which leads to:

$$\min_{i \neq k} a^{(i)} b^{(i)} \geq -\sqrt{\|a\|_2^2 \|b\|_2^2 - x^2}. \quad (65)$$

□

LEMMA E.8.

Let $D = \text{diag}(d_1, d_2, \dots, d_n)$ be a diagonal matrix with strictly positive entries, i.e., $d_i > 0$ for all $i \in \{1, 2, \dots, n\}$. Then, for any vector $b \in \mathbb{R}^n$, the following inequality holds:

$$\|Db\|_2 \leq \left(\max_i d_i \right) \cdot \|b\|_2. \quad (66)$$

PROOF.

Let $b = [b_1, b_2, \dots, b_n]^T$. Then,

$$Db = [d_1 b_1, d_2 b_2, \dots, d_n b_n]^T, \quad (67)$$

and

$$\|Db\|_2 = \sqrt{\sum_{i=1}^n (d_i b_i)^2} \leq \sqrt{\sum_{i=1}^n (\max_j d_j)^2 (b_j)^2} = (\max_i d_i) \sqrt{\sum_{i=1}^n (b_i)^2} = (\max_i d_i) \|b\|_2 \quad (68)$$

□

F Additional Experiments on Parameter Analysis

We perform the same analysis of the four factors on the Epinions and synthetic datasets, and the results show a high degree of consistency with those on the Ciao dataset.

F.1 Analysis of α , β , γ and ϵ on Epinions Dataset

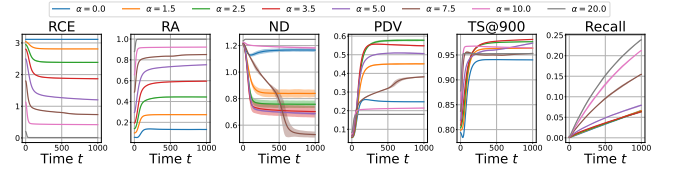


Figure 13: Metric trends over time under varying α on Epinions dataset.

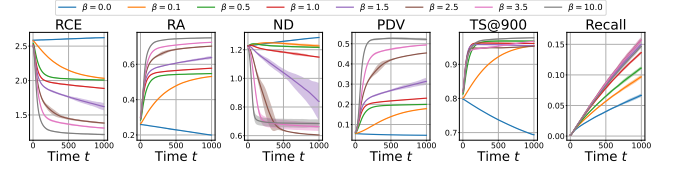


Figure 14: Metric trends over time under varying β on Epinions dataset.

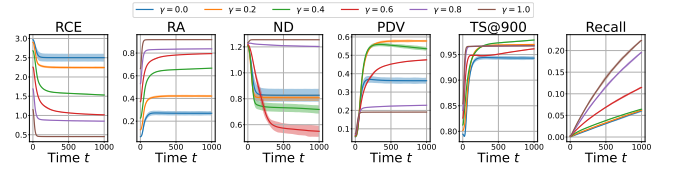


Figure 15: Metric trends over time under varying γ on Epinions dataset.

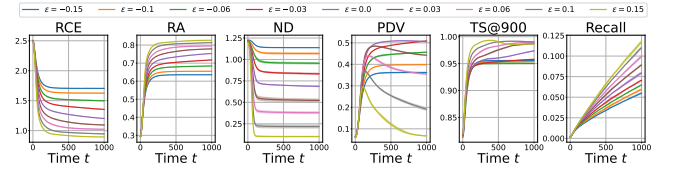


Figure 16: Metric trends over time under varying ϵ on Epinions dataset.

F.2 Analysis of α , β , γ and ϵ on Synthetic Dataset

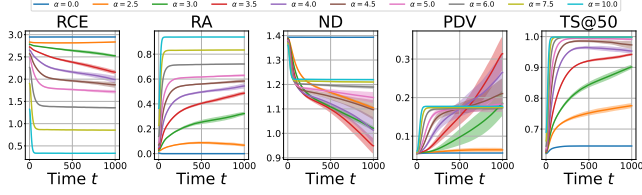


Figure 17: Metric trends over time under varying α on Synthetic dataset.

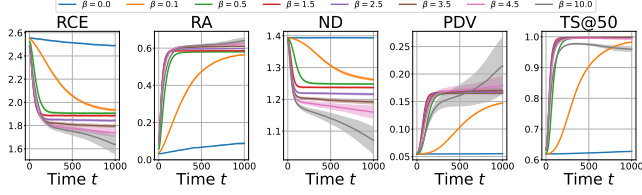


Figure 18: Metric trends over time under varying β on Synthetic dataset.

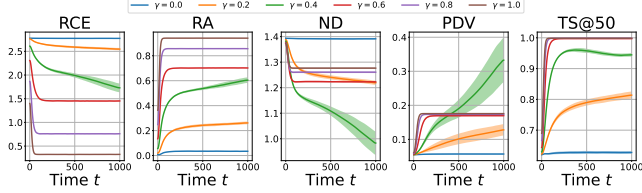


Figure 19: Metric trends over time under varying γ on Synthetic dataset.

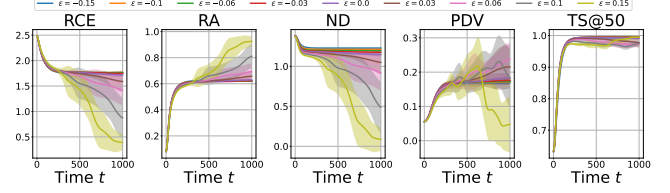


Figure 20: Metric trends over time under varying ϵ on Synthetic dataset.

Table 4: Parameter sensitivity analysis of four strategies on Ciao dataset.

	Ciao	RCE↑	RA↑	ND↑	PDV↓	TS@300↓
UA α	$\alpha = 1$	1.21±0.004	0.74±0.001	0.90±0.005	0.30±0.002	0.96±0.001
	$\alpha = 5$	1.24±0.003	0.73±0.001	0.94±0.001	0.29±0.002	0.96±0.001
	$\alpha = 10$	1.28±0.000	0.72±0.000	0.95±0.002	0.28±0.002	0.96±0.001
	$\alpha = 20$	1.32±0.004	0.71±0.001	0.98±0.001	0.27±0.000	0.95±0.000
FUA	$\rho = -0.01$	1.16±0.004	0.75±0.001	0.86±0.004	0.32±0.003	0.96±0.001
	$\rho = 0.02$	1.27±0.009	0.73±0.002	0.92±0.007	0.29±0.005	0.96±0.000
	$\rho = 0.05$	1.36±0.005	0.70±0.001	0.96±0.003	0.27±0.002	0.95±0.000
	$\rho = 0.08$	1.44±0.004	0.68±0.001	1.00±0.001	0.26±0.002	0.95±0.001
DPP	$\theta = 0.500$	1.13±0.001	0.78±0.000	0.89±0.006	0.30±0.003	0.96±0.000
	$\theta = 0.501$	1.38±0.001	0.74±0.000	0.97±0.004	0.27±0.002	0.95±0.000
	$\theta = 0.502$	1.61±0.001	0.70±0.000	1.04±0.003	0.24±0.003	0.94±0.001
	$\theta = 0.503$	1.80±0.001	0.66±0.000	1.09±0.002	0.22±0.001	0.94±0.001
SAR	$\omega = 100$	1.22±0.005	0.72±0.001	0.85±0.016	0.31±0.004	0.96±0.002
	$\omega = 1000$	1.26±0.009	0.68±0.003	0.94±0.008	0.28±0.004	0.95±0.001
	$\omega = 2000$	1.24±0.003	0.69±0.001	0.90±0.019	0.29±0.005	0.95±0.000
	$\omega = 5000$	1.25±0.004	0.68±0.001	0.92±0.013	0.28±0.004	0.95±0.001

G Additional Experiments with Mitigation Strategies on Ciao Dataset

Through further parameter tuning of the four mitigation strategies on the Ciao dataset in Table 4, we observe that sacrificing a certain degree of accuracy can lead to more effective alleviation of echo chambers and user homogenization.